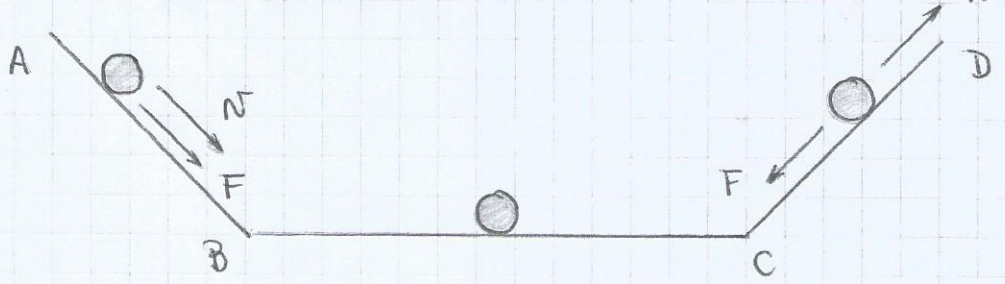
Unidad II: Dinámica

Interacción = fuerza, la acción ejercida sobre un sistema que modifica su estado de movimiento (F)

Un cuerpo aislado o libre es un cuerpo sobre el cual no actúa ninguna fuerza.

$$\sum_{k} F_k = 0$$

Galileo - Galilei:

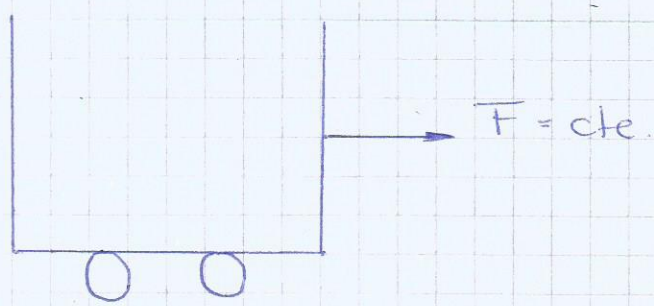


- 1° Principio de inercia: Todo cuerpo (sistema) libre de interacción se mueve con M.R.U o mantiene su estado de reposo.

$$\sum_{k} F_k = 0 \begin{cases} v = 0 \\ v = cte \end{cases}$$

$F = k . a \quad (k > 0)$

$F \parallel a$



masa: m de la inercia del cuerpo, la resistencia de los cuerpos al cambio de movimiento.

$m_1 \longrightarrow a_1$

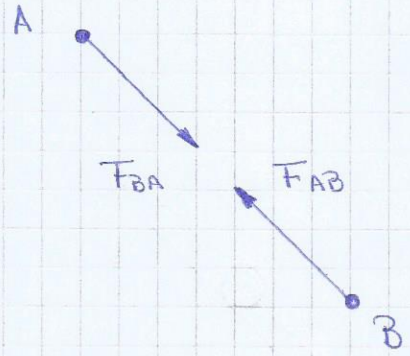
$m_2 \longrightarrow a_2$

$$\frac{m_1}{m_2} = \frac{a_2}{a_1}$$

- 2° Principio de masa:

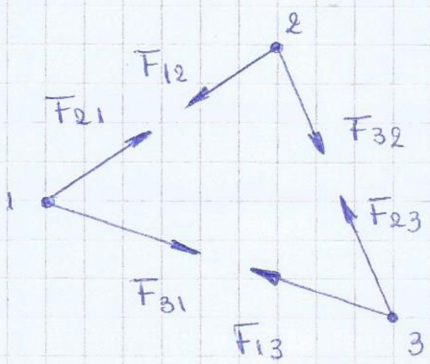
$$\sum_K F_K = m \cdot a$$

- 3° Principio de acción y reacción.



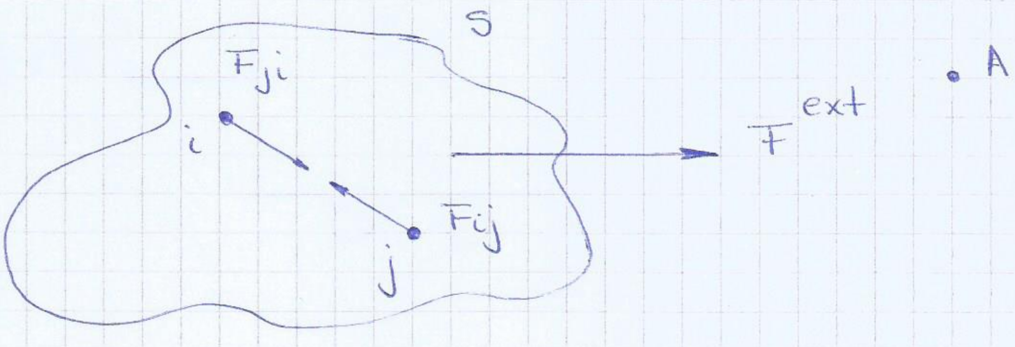
$$F_{AB} = -F_{BA}$$

se ejercen de a pares.



$$a_1 = \frac{1}{m_1}(F_{21} + F_{31})$$

Fuerzas internas y externas:



Fuerza interna: tiene reacción dentro del sistema

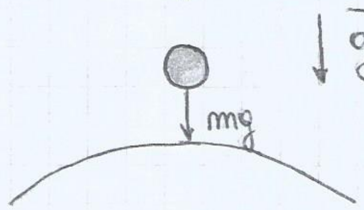
Fuerza externa: tiene reacción fuera del sistema.

$$\sum_k F_k = F^{ext} + \sum_{i \neq j} f_{ij} = m\,a$$

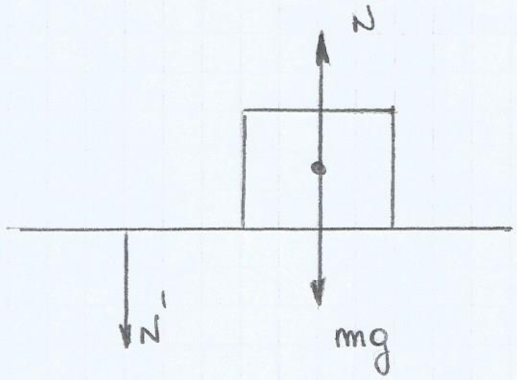
$$F^{ext} + \sum_{i>j} (f_{ij} + f_{ji}) = F^{ext} = m\,a$$

Fuerza peso: $P = m \cdot a$ con $a = g$

$P = m\,g$



Fuerzas de contacto: N' es la fuerza que m hace sobre la mesa

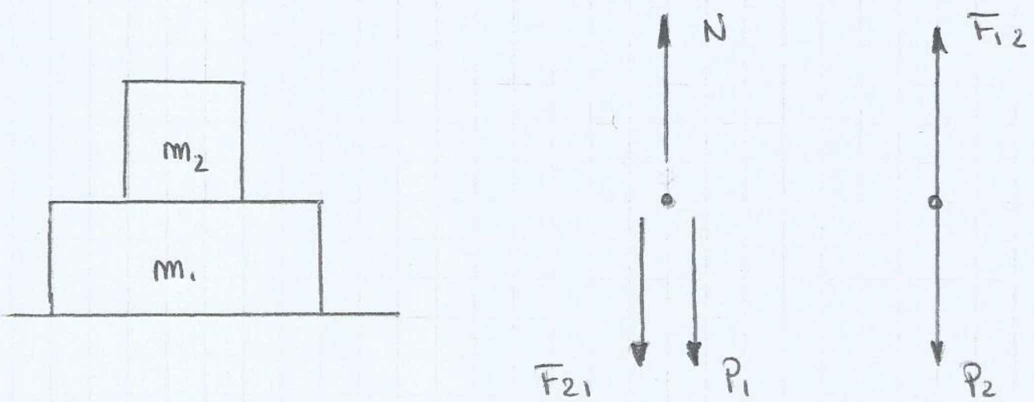


$|F| = |N|$

$F = -N$

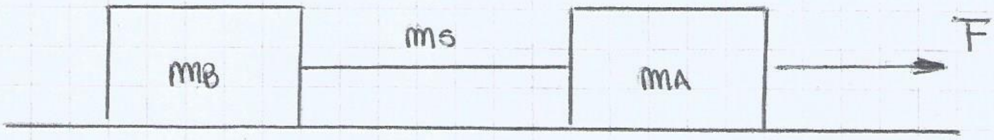
$P + N = m\,a = 0$

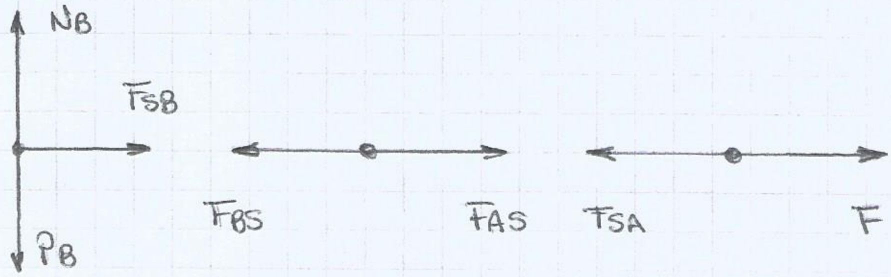
$P = N$



$Fc_{12} + P_2 = m_2 a_2 = 0$

$Fc_{21} + P_1 + N = m_1 a_1$





$F_{SB} = m_B a_B \quad (1) \qquad F_{SB} = m_B a_B$

$F_{BS} + F_{AS} = m_S a_S \quad (2) \qquad -F_{BS} + F_{AS} = m_S a_S$

$F_{SA} + F = m_A a_A \quad (3) \qquad F - F_{SA} = m_A a_A$

Como $m_S \simeq 0 \Rightarrow (2)\ -F_{BS} + F_{AS} = 0$

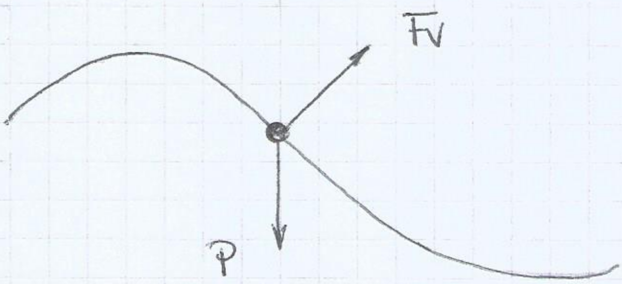
$F_{AS} = F_{BS} = T$

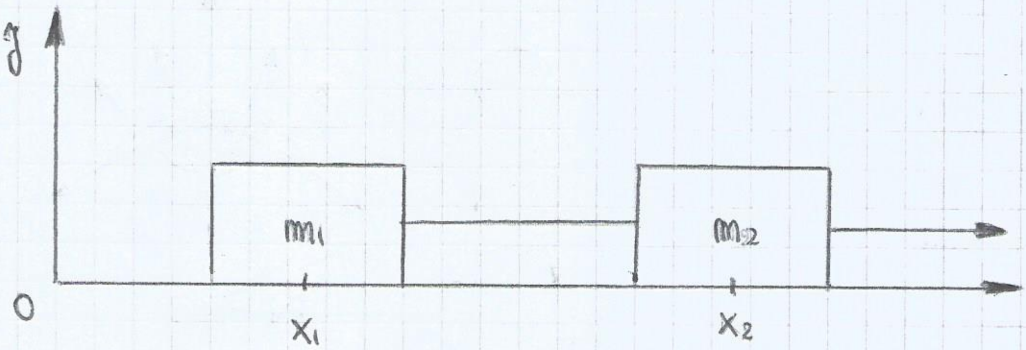
$T = m_B a_B$

$-T + F = m_A a_A$

Condiciones de vínculos:

Son condiciones que limitan el movimiento.





Vínculos: $l = x_2 - x_1$

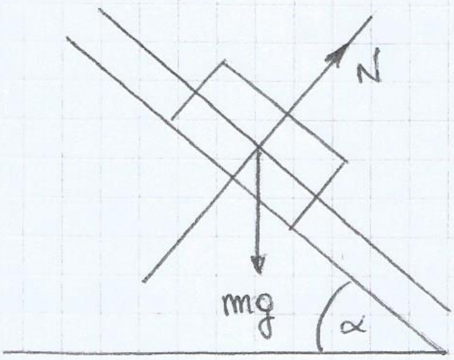
$$\ddot{l} = \ddot{x}_2 - \ddot{x}_1$$

$$0 = \ddot{x}_2 - \ddot{x}_1$$

$$\ddot{x}_2 = \ddot{x}_1$$

Unidades: M.K.S C.g.S

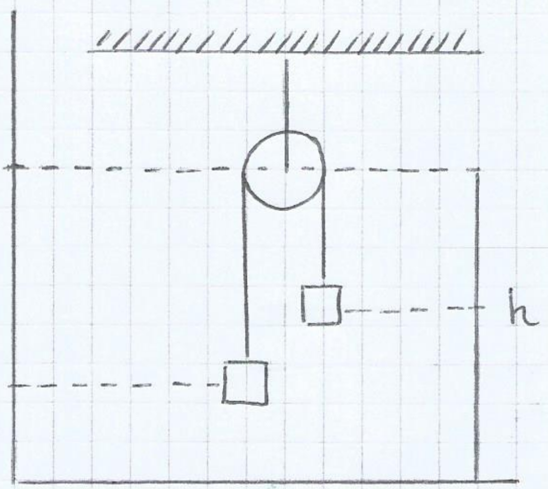
$[F] = [m][a]$ $\quad Kg \cdot \frac{m}{s^2} = N$ $\quad g \cdot \frac{cm}{s^2} = dyn$

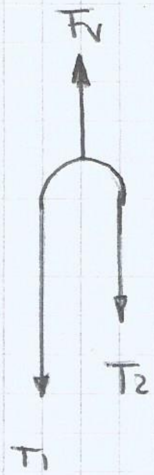


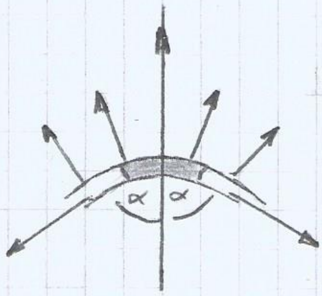
$\hat{x}: mg \, sen \, \alpha = m \, a_x$

$\hat{y}: N - mg \cos \alpha = m \, a_y$

1) Identificar el sistema
2) Identificar F^{ext}
3) Hacer un D.C.L
4) Elegir un sistema de coordenadas
5) Escribir el segundo principio de la dinámica.
6) Identificar la condición de vínculo.







$$\bar{F_V} = 2T$$

$$\hat{j}: F_V - T_1 \cos\alpha - T_2 \cos\alpha = 0$$
$$\hat{x}: -T_1 \operatorname{sen}\alpha + T_2 \operatorname{sen}\alpha = 0$$

$$T_1 = T_2 = T$$

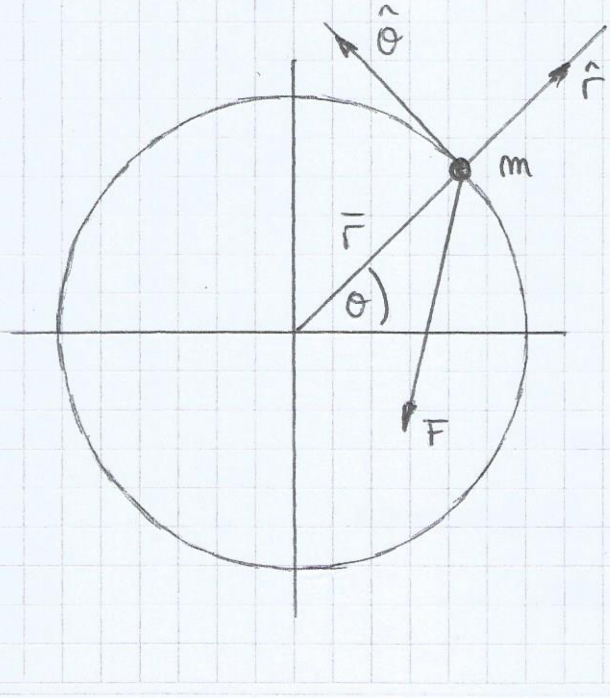
Condiciones de vínculo

$$l = h - x_1 + x_2$$
$$\ddot{l} = \ddot{h} - \ddot{x}_1 + \ddot{x}_2$$
$$0 = 0 - \ddot{x}_1 + \ddot{x}_2$$
$$\ddot{x}_1 = \ddot{x}_2 = a$$

$$\begin{cases} -P_1 + T = m_1 \ddot{x}_1 \\ P_2 - T = m_2 \ddot{x}_2 \end{cases}$$

$$\begin{cases} -m_1 g + T = m_1 a \\ m_2 g - T = m_2 a \end{cases}$$

Fuerzas en movimientos circulares:

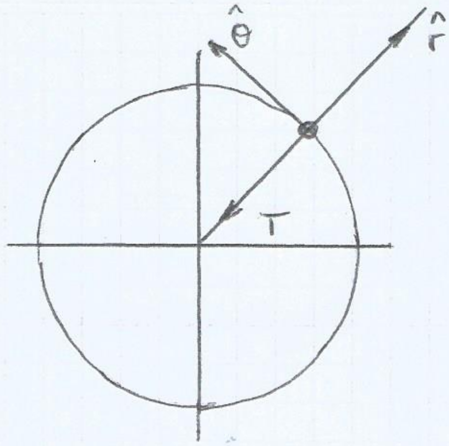


Condición de vínculo: $r = R = cte$

$$\hat{r}: -F_{\hat{r}} = -F \operatorname{sen}\theta = (\ddot{r} - r\dot{\theta}^2)m$$
$$\hat{\theta}: -F_{\hat{\theta}} = -F\cos\theta = (r\ddot{\theta} + 2\dot{r}\dot{\theta})m$$

$$\hat{r}: -F \operatorname{sen}\theta = -mR\dot{\theta}^2$$
$$\hat{\theta}: -F\cos\theta = mR\ddot{\theta}$$

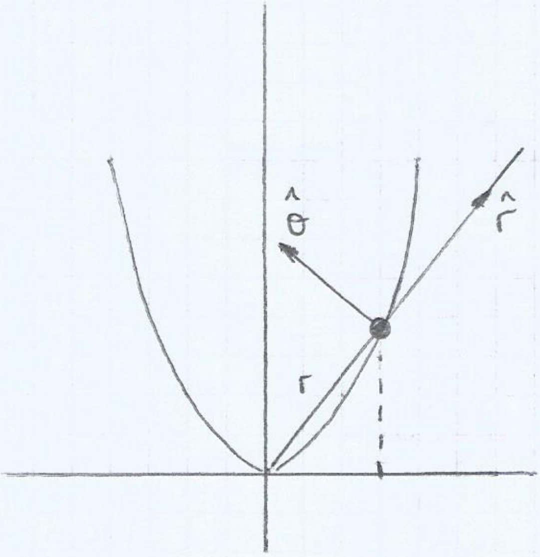
NOTA



Condición de vínculo: $r = R = cte$

$\hat{r}: \; -T = -m r \dot{\theta}^2$

$\hat{\theta}: \; 0 = m r \ddot{\theta} \Rightarrow \ddot{\theta} = 0 \quad \dot{\theta} = cte$



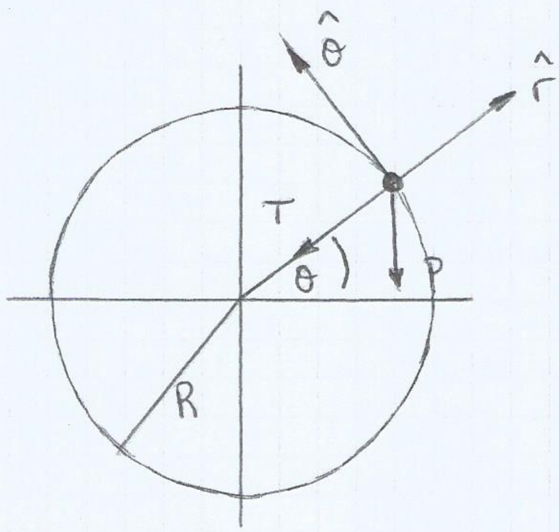
$\hat{r}: \; -mg\,sen\theta + F_v = m\ddot{r} - m r \dot{\theta}^2$

$\hat{\theta}: \; -mg\cos\theta + F_v = m r \ddot{\theta} + 2m\dot{r}\dot{\theta}$

Condición de vínculo:

$$\left.\begin{array}{l} y = a x^2 \\ y = r\,sen\theta \\ x = r\cos\theta \end{array}\right\} \begin{array}{l} r\,sen\theta = r^2 \cos^2\theta \, a \\ \\ r(\theta) = \dfrac{sen\theta}{\cos^2\theta \, a} \end{array}$$

Ejemplo:



$\hat{r}: \; -T - mg\,sen\theta = -m\dot{\theta}^2 R$

$\hat{\theta}: \; -mg\cos\theta = m R \ddot{\theta}$

$$-mg\cos\theta = m R \frac{d\dot{\theta}}{d\theta}\frac{d\theta}{dt}$$

$$-g\cos\theta = R \frac{d\dot{\theta}}{d\theta}\dot{\theta}$$

$$-g\cos\theta \, d\theta = R\dot{\theta}\, d\dot{\theta}$$

$$-g\int_0^{\theta'} \cos\theta \, d\theta = R\int_{\dot{\theta}_0}^{\dot{\theta}'} \dot{\theta}\, d\dot{\theta}$$

$$-g \operatorname{sen} \theta = \left(\frac{\dot{\theta}^2}{2} - \frac{\dot{\theta}_0^2}{2} \right) R$$

$$\dot{\theta} = \sqrt{\dot{\theta}_0^2 - 2 \frac{g}{R} \operatorname{Sen} \theta}$$

$\operatorname{Sen} \theta_R = \frac{R \omega_0^2}{2g} < 1 \qquad \omega_0^2 < \frac{2g}{R} \qquad \theta_R$, ángulo de retorno

Para que posición la tensión vale cero.

$$-m g \operatorname{sen} \theta_v = -m R \dot{\theta}^2$$

$$\dot{\theta}^2(\theta_v) = -\frac{g \operatorname{sen} \theta_v}{R} = \omega_0^2 - \frac{2g \operatorname{sen} \theta_v}{R}$$

$$\operatorname{Sen} \theta_R > \operatorname{Sen} \theta_v \begin{cases} \omega_0^2 = \dfrac{3g \operatorname{sen} \theta_v}{R} \\ \\ \operatorname{Sen} \theta_v = \dfrac{\omega_0^2 R}{3g} \end{cases}$$

Condición de equilibrio:

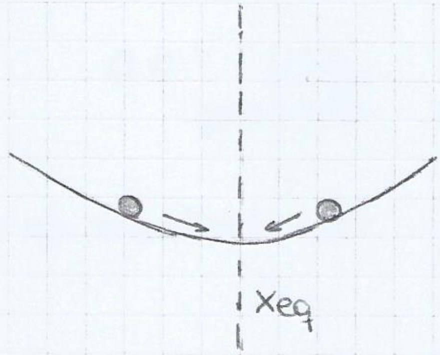
Equilibrio: $a = 0$ $\qquad \sum_k F_k^{ext} = 0$

Reposo: $v = 0$

I) equilibrio estable

II) equilibrio inestable

III) equilibrio indiferente.

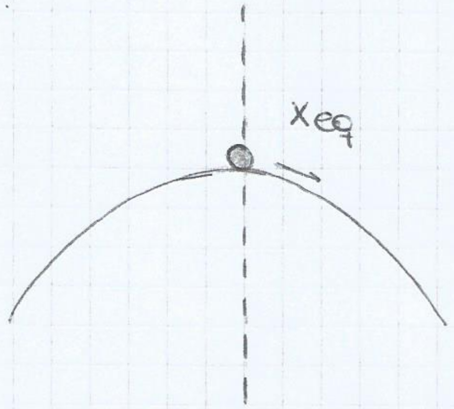


equilibrio estable

x_{eq} , posición de equilibrio

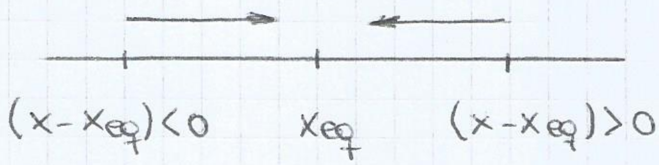
$$F(x_{eq}) = 0$$

$$F(x) = F(x_{eq}) + \left.\frac{dF}{dx}\right|_{x_{eq}} (x - x_{eq})$$



equilibrio inestable

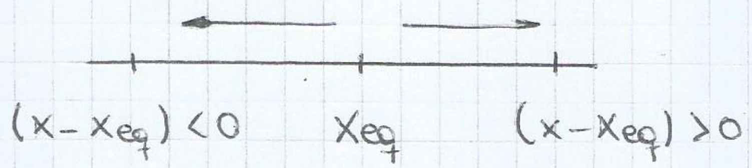
equilibrio indiferente



Si eq es estable

$$\left.\frac{dF}{dx}\right|_{x_{eq}} < 0$$

$\forall x$ alrededor de x_{eq}



Si eq es inestable

$$\left.\frac{dF}{dx}\right|_{x_{eq}} > 0$$

$\forall x$ alrededor de x_{eq}

Si el equilibrio es indiferente

$$\left.\frac{d^n F}{dx^n}\right|_{x_{eq}} = 0 \qquad \forall n \in \mathbb{N}$$

NOTA